



Mission and system design for the EarthNext Cubesat VLEO mission

Tobia Armando La Marca¹ · Maria Daniela Graziano² · Michele Grassi² · Lorenzo Iannascoli³ · Leonardo Cavicchiolo³ · Annalisa Capanni³ · Matteo Duzzi⁴ · Giovanni Maria Capuano⁵ · Vito Fortunato⁶ · Christian Cardenio⁷ · Giuseppe Leccese⁸ · Marianna Rinaldi⁸ · Marco Di Clemente⁸ · Roberto Luciani⁸ · Silvia Natalucci⁸ · Vincenzo Pulcino⁸

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Abstract

EarthNext is a CubeSat mission developed within the ALCOR program, an initiative supported by the Italian Space Agency (ASI) to develop innovative CubeSat missions addressing diverse space-related challenges. Specifically, the EarthNext mission is designed for the In Orbit Demonstration (IOD) of operating a CubeSat in Very Low Earth Orbit (VLEO). Operating in such a low-altitude environment is of particular interest due to the unique advantages offered by VLEO, including enhanced spatial resolution and radiometric performance in Earth Observation (EO), reduced revisit time, and the possibility of using low-cost, volume-limited platforms. Consistently, EarthNext also aims to deliver value-added products and services for land and marine applications through the acquisition of high-resolution multispectral images of customer-selected targets over Italy. As a result, the platform, conceived as a 16U CubeSat, has been designed to excel in sustained EO tasks and autonomous operations since it can rely on fundamental enabling technologies such as an innovative electric Propulsion Subsystem (PS), a dedicated advanced optical payload, and on-board Artificial Intelligence. The following contribution will then focus on the mission design of EarthNext, presenting the main outcomes of the mission analysis and an overview of the mission's concept of operations. The contribution will be further enhanced by including the most recent advancements in platform design to achieve the mission's objectives.

Keywords CubeSat · Very low earth orbit · Earth observation · Autonomous operations

1 Introduction

EarthNext is a CubeSat mission belonging to the ALCOR program, a CubeSat initiative conceived and funded by the Agenzia Spaziale Italiana (ASI), with the objective of

fostering the development of innovative CubeSat missions and high-level enabling technologies for several space-related applications [1]. The EarthNext mission has been developed with the primary objective to demonstrate the feasibility of operating with a small platform in the VLEO environment for a timespan of at least 3 years. Supervised and funded by ASI, the project involves a team led by Officina Stellare, the prime contractor, and including AIKO Space, Planetek Italia, T4I—Technology for Propulsion and Innovation, TSD-Space, and the Department of Industrial Engineering of the University of Naples Federico II. The project has recently completed its Preliminary Design Review, and the subsequent mission design phases C/D/E1 have already been approved.

Although the scientific community is still debating a coherent definition, the term VLEO typically refers to orbits with a mean altitude of less than 450 km [2]. These orbits have garnered significant attention over the past decades due to their advantages for scientific and industrial applications which are not limited to individual satellites but can be

✉ Tobia Armando La Marca
tobiaarmando.lamarca-ssm@unina.it

¹ Scuola Superiore Meridionale, Largo San Marcellino 10, 80138 Naples, Italy

² Department of Aerospace Engineering, University of Naples Federico II, Piazzale Tecchio 80, 80125 Napoli, Italy

³ Officina Stellare S.P.A, Sarcedo, Italy

⁴ T4I—Technology for Propulsion and Innovation, Monselice, Italy

⁵ TSD-Space, Pozzuoli, Italy

⁶ Planetek Italia, Bari, Italy

⁷ AIKO Space, Torino, Italy

⁸ Agenzia Spaziale Italiana (ASI), Roma, Italy

framed and integrated into broader concepts, such as constellations delivering EO or Space Situational Awareness services. Over the last years, several missions have successfully operated in a similar orbital regime [3–6] and other missions are programmed for the near future [7], definitively highlighting the VLEO's potential for advancing both scientific research and technological innovation.

The benefits of such orbits are particularly evident for EO missions. Indeed, the reduced altitude enhances spatial resolution and radiometric performance, enabling the use of compact payloads compatible with the typical CubeSat volume constraints [8]. The relatively simple spacecraft design and realization, as well as the limited effort required to put the spacecraft into operational orbits, reduce the platform development and launch costs [9]. At the same time, the proximity to Earth decreases telecommunications latency. Additionally, the VLEO environment is naturally predisposed to the absence of space debris, facilitating also the spacecraft disposal due to the high orbital decay rate.

However, despite these advantages, VLEOs present notable technical challenges that must be addressed during mission planning and platform development. The high decay rate induces the need for the integration of a PS. Furthermore, the abundance atomic oxygen, combined with the high orbital velocity, causes a high corrosion of the external coatings [10], which must be mitigated with an appropriate selection of the spacecraft materials. Moreover, the high ground-relative velocity also requires tailored attitude control maneuvers to reduce image blurring and maximize exposure times. As a result, a VLEO spacecraft requires the development of a GNC subsystem that is responsive, agile and at the same time robust to the external environmental perturbations not easily predictable.

VLEOs also offer good protection against radiation. In the already mentioned study [8], using the SPENVIS software [11], whose application is also referenced by the ECSS guidelines [12], a comparison of proton and electron fluxes at altitudes of 300 km and 600 km is presented. The study highlights that, while the peak electron flux at 300 km does not significantly decrease, the geographical extent of regions with high flux levels is considerably reduced. Conversely, the trapped proton flux within the South Atlantic Anomaly (SAA) is lower at 300 km compared to 600 km, primarily due to the increased atmospheric density at lower altitudes. However, since most EO missions, including EarthNext, typically adopt Sun-Synchronous Orbits (SSOs), this implies potentially multiple daily passages through high-radiation regions such as the polar horns and the SAA, where elevated fluxes of electrons, protons, and galactic cosmic rays are encountered [13]. Consequently, despite the inherent shielding benefits of VLEO, a comprehensive radiation analysis remains necessary to ensure that the total accumulated dose remains within tolerable limits for the spacecraft's COTS

hardware. Furthermore, the tight scheduling of operations performed both for the in-orbit sustainment of the spacecraft and for the completion of the mission's scientific objectives, calls for the development of the platform autonomous capabilities to optimize, and eventually re-plan, the mission operation schedule, which cannot ignore online predictions about the external environment and the platform status. These requirements can be addressed through AI-driven solutions designed to enhance payload utilization, streamline mission operations, and data handling. For EO missions, such solutions can also integrate weather prediction and cloud cover observations over Area of Interest (AOI) to further refine observation planning.

To tackle these VLEO-specific challenges, the EarthNext consortium identified and mitigated critical issues from the early phases of the mission design process. These efforts are the backbone of this work, which provides an overview of the EarthNext mission concept and the latest updates on its design. The paper is structured as follows: Sect. 2 outlines the EarthNext mission objectives and the rationale behind the chosen nominal orbit. Section 3 details the spacecraft configuration and the subsystems design, while in Sect. 4 the concept of operations and GNC modes of EarthNext have been described. Finally, conclusions are drawn in Sect. 5.

2 Mission objectives and orbit

The main mission objective of EarthNext is to demonstrate the possibility of continuously operating a CubeSat in VLEO at predefined altitudes. However, the orbital parameters for the mission have mainly been defined based on the scientific objectives. Indeed, the mission also aims:

- To produce high-value products, i.e., high spatial resolution multispectral images, for land and marine applications over specific customer-based AOIs;
- To create a synergistic dataset with higher-performance multispectral missions (using as baseline for a preliminary design Sentinel-2 satellites) while also considering others EO passive remote sensing infrastructures;
- To exploit a secondary camera for cloud detection over the AOI;
- To enhance the autonomous operations planning by adopting an on-board, AI-based algorithms.

Stemming from the mission objectives, the mean equatorial altitude of the orbit has been nominally set to 302 km, i.e., the mean Semi-Major Axis (SMA) has been defined as $SMA = 6680.2$ km, which allows the orbit to be classified as VLEO while ensuring a repetition factor of $Q = 159/10$, compatible with the one of Sentinel-2 satellites for the synergistic dataset creation. The inclination (INC) was instead

set to obtain an SSO, which simplify the configuration of the solar panels for the electrical power generation, a key aspect due to the high energy consumption expected during the operations. In particular, the nominal inclination selected is $INC = 96.68^\circ$, with the eccentricity (ECC) set to $ECC = 0.0011$, to realize a frozen orbit and nullify the long periodic variations of the Argument of Perigee (AOP). The AOI for a preliminary orbit assessment has been defined within the longitude range of 5°E to 20°E and the latitude range of 35°N to 48°N , based on a WGS84 ellipsoid. The AOI has been depicted for clarity in Fig. 1.

To guarantee good illumination conditions when the spacecraft overflies the AOI, the Local Time of Descending Node (LTDN) has been defined equal to 10:30 AM. It further allows to avoid the morning haze [14], thus reducing the probability of cloud coverage, and it is particularly suitable for production agriculture as the morning fog tends to lift between 9:00 AM and 13:00 AM, letting the plants reach their normal metabolic state. Consequently, the LTDN has been constrained to be between 10:00 AM and 12:00 AM for the whole operational lifetime. The nominal mission lifetime is set to three years. Nonetheless, all subsystems of the space segment have been designed and properly sized to support a total operational lifespan of up to five years, thus enabling a potential mission extension beyond the baseline duration. Orbital simulations carried out in a high-fidelity simulation environment have, however, highlighted that the solar gravitation may compromise the desired orbital configuration, by progressively decreasing the inclination and not allowing the LTDN to be maintained within the mission requirements. The LTDN/LTAN drift is generally counteracted through the adoption of out-of-plane maneuvers. A preliminary strategy based on the natural dynamics of the satellite has been considered to avoid the overloading of the propulsion system,

already burdened by numerous drag compensation working cycles. The basic idea is to insert the spacecraft into an orbit with a biased value of inclination which compensates for the drift accumulated by EarthNext during its operational life while still preserving the features of a quasi-SSO. For a preliminary evaluation, a first refinement on the inclination has been performed to also account for the gravitational perturbation up to the J4 harmonic. By exploiting the mean Kozai-Izsak orbital-elements formulation reported in [15], the INC has been slightly modified to 96.702° . Then, a preliminary estimation of the inclination drift for the nominal SSO has been computed by considering the simplified equation by Chobotov in [16], which estimated a $-0.33^\circ/\text{year}$ drift in INC, leading to a final inclination of 96.8° to compensate for such a drift in 3 years. In Fig. 2, the LTDN trend over a 5-year time-span is shown for both the biased and nominal orbits, clearly highlighting that the required inclination bias allows compliance with the LTDN constraints, albeit over a longer period. However, more precise estimations will be provided in the future, which could lead to a reduced inclination bias.

Although this orbit is no longer strictly an SSO, subsequent analyses carried out by the consortium have demonstrated the possibility of compliance with the electrical power requirements. In Table 1, the final orbit parameters selected for EarthNext mission are reported. The Right Ascension of the Ascending Node (RAAN) has been computed assuming the 1st January 2026 13:00:00.000 UTC-G as the reference epoch. Based on these parameters, and incorporating in the simulations the drag make-up maneuver strategy designed for EarthNext—further detailed in Sects. 3.2 and 4.1—the mean revisit time in 3 years of operations, over the AOI and under sun-illuminated conditions is of about 27 h and 40 min.

Concerning the relation between the orbit and the ground-station service capabilities, at an early stage of Phase B, the consortium identified LeafSpace as a potential ground station service provider. The analysis of ground station contact opportunities was based on realistic orbit propagation, considering environmental perturbations and the drag compensation strategy adopted for EarthNext, over a 10-day interval, chosen to coincide with the nominal orbit's ground-track repetition cycle. These results were validated in close collaboration with LeafSpace. For payload data downlink via X-band, only contact windows longer than 2 min and not overlapping with Propulsion Subsystem Activation (PSA) phases or Acquisition Phases (APs) were retained. The 2-min minimum ensures the transfer of approximately 50% of the acquired data in the unlikely Worst-Case (WC) scenario of an 8-min observation segment (see SubSect. 3.5 for further details). Telemetry, Tracking & Commands (TT&C) rely on both S-band and UHF-band ground stations, selected from LeafSpace's candidates list. Imposing a minimum



Fig. 1 AOI for EarthNext mission

Fig. 2 LTDN temporal variation during the expected EarthNext lifespan

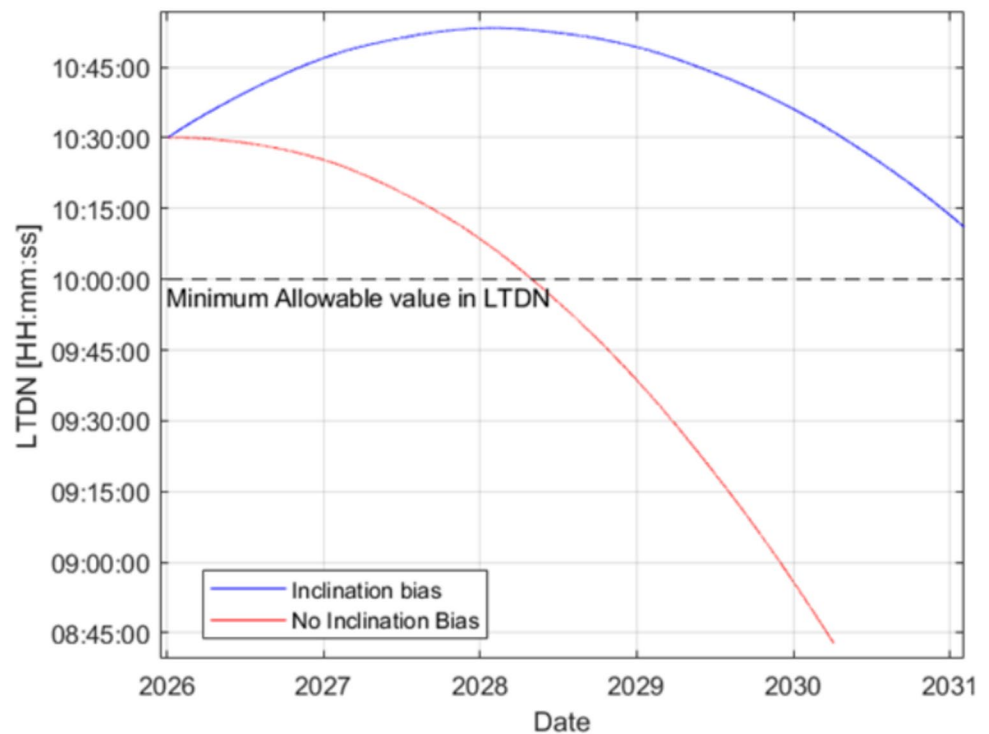


Table 1 EarthNext orbit parameters

Parameter	Value
SMA (km)	6680.2
ECC	0.011
INC (°)	96.8
LTDN (hh:mm)	10:30
RAAN (°)	78.7
AOP (°)	90.0

elevation angle constraint of 10° yielded to an X-band mean contact duration of approximately 4 min, with enough contact opportunities between two consecutive overflights of the AOI. Similarly, for S-band, supported by most of the same ground stations as the X-band, the average contact duration remained approximately 4 min, allowing for an adequate number of GS contacts within two subsequent overflights. For the UHF-band, even when accounting for possible communications during the PSAs, the average number of contacts was significantly lower, due to the reduced number of GSs providing such service. The analysis then indicated that relying solely on UHF for TT&C could result in cases where no ground station contact is available during PSA phases. The inclusion of S-band communications would significantly improve TT&C coverage during such phases and enhances mission robustness.

Finally, the decision to adopt a side-mounted solar array configuration is motivated by both mission analysis considerations and the need to ensure efficient battery recharging.

From a mission analysis perspective, the fixed orientation of a side-mounted array in nadir attitude yields an almost constant power profile during sunlight, with a peak when the front-facing panel is illuminated. The sizing of the side-mounted array was therefore based on the following expression for the n -th panel:

$$P_n = P_{mp} \nu \cos(\theta) \eta_{uv} \eta_{tc} \eta_{ef} \quad (1)$$

where

- P_{mp} is the power output of each solar panel, conservatively specified by the prospective supplier using the end-of-life (EOL) value at a working-cell temperature (WCT) of 70°C ;
- ν is an eclipse coefficient ($\nu = 1$ in sunlight, $\nu = 0$ in Earth's shadow);
- θ is the angle between the Sun-satellite direction and the normal to the solar panel (assuming that power generation ceases beyond an 85° threshold);
- η_{uv} is an efficiency parameter that consider the ultraviolet-ray effects;
- η_{tc} is an efficiency parameter that considers thermal-cycling effects;
- η_{ef} is an efficiency parameter that considers the time-degradation effects;

Using the reported mathematical relationship, a parametric WC analysis was conducted to determine the number of

panels in the side-mounted solar array that, together with the body-mounted solar panels, satisfy the satellite's power requirements (see Sect. 3 for further details about the EPS as well as the platform configuration and representation). Additionally, the side-mounted configuration ensures that, during Forward Motion Compensation (FMC) maneuvers (primarily intended as rotations around the pitch axis), the spacecraft wet area does not significantly increase due to the lateral solar array itself, thus avoiding a substantial worsening of the attitude control effort.

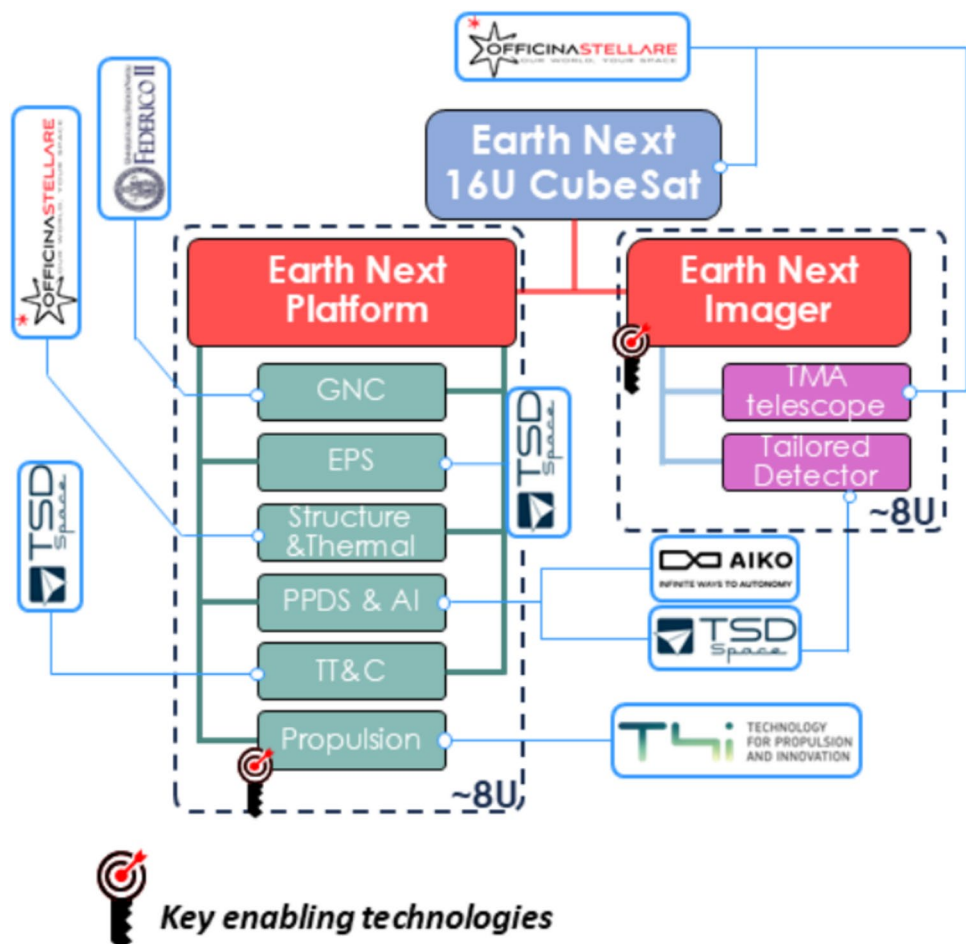
3 EarthNext space platform architecture

The EarthNext platform has been conceived as a 16U CubeSat, allocating 8U for the Imager and 8U for all the other subsystems required for spacecraft operability. For the sake of clarity, a schematic of the platform's subsystems is shown in Fig. 3.

There are two key enabling technologies. The first is the Three-Mirror Anastigmatic (TMA) telescope, designed and manufactured by Officina Stellare, together with a dedicated detector designed and realized by TSD, to capture

high resolution and quality images of specific target locations. The latter is an integrated electric PS, tailored for VLEO and provided by T4I, capable of performing repetitive drag-compensation maneuvers. The Electrical Power Subsystem (EPS) has been designed to meet the spacecraft power demand, which peaks at 84 W during thruster activation. The bus power is unregulated, oscillating between 11.8 and 16.8 V. On the contrary, the GNC subsystem has been designed to guarantee a 3-axis stabilization of the platform to accomplish all the planned attitude maneuvers with the desired accuracy, as better detailed in Sects. 3.3 and 4.1. The external coatings selection process has been tested and verified, based on both the Material4Cubesat test campaign provided by ESA and consortium analysis using the ESABASE2 software [17]. In particular, the atmospheric species concentrations were modeled using the NRLMSISE-00 atmospheric model, recommended by the ECSS [12]. Solar activity conditions were set to WC 2000–2002 triennium, with a daily F10.7 index of 280 SFU and a 3-month average of 180 SFU, while the geomagnetic activity index A_p was fixed at 165 according to the WC condition suggested in [18]. The effects of Atomic Oxygen (AO) particle jitter due to thermal motion were also included in the model. The WC analysis

Fig. 3 Schematic of the EarthNext subsystems



estimation highlighted the AO max flux on the spacecraft surfaces directly exposed to the velocity vector direction, i.e., the frontal part of the spacecraft and the edges of the solar array. The analysis confirms that, although the high AO flux, the erosion depth for the materials selected was not a problem, but attention should be paid to the inclusion of polymeric materials over the spacecraft, since suffering of high erosion yield. At the same time, the Material4Cubesat test campaigns were performed on the material selected for the optical coating and the thermal coating and confirmed that no critical degradation was recorded for the tested flux. As noted in the previous subsection, TT&C uses S-band and UHF for telemetry, tracking, and command, while payload data transmission is handled mainly via X-band. Finally, an AI-based internal logic has been designed to implement the online replanning of the operations, as well as an optimization of the data acquisition and data handling. A pictorial front-view of the spacecraft with its main components is provided in Fig. 4, together with the orbital reference frame, referred to as the Local Vertical Local Horizontal (LVLH) reference frame. In this frame, the x_{LVLH} point toward the spacecraft direction of motion, z_{LVLH} points toward the nadir

and, the y_{LVLH} completes the triad. The main subsystems are further described in the following subsections.

3.1 EarthNext imaging system

The small volume available drove the design of the Earth-Next Imager subsystem toward an unobscured, mirror-based solution to both fit into 8U and maximize the modulation transfer function. A three-mirror anastigmatic (TMA) architecture has been selected, depicted for clarity in Fig. 5. Such typology of telescopes is generally corrected for spherical aberration, coma and astigmatism. The telescope Field Of View (FOV), expressed in across-track (ACT) x along track (ALT) is $3^\circ \times 1.5^\circ$, allowing to reach an ACT swath width of 17 km.

The imagery system allows finally to get a 3.9 m Ground Sampling Distance (GSD) in nadir pointing. The main design characteristics of the TMA telescope have been finally reported in Table 2.

The electro-optical payload relies on a custom multi-spectral filter array implementing a spatio-spectral scanning acquisition technique. A multispectral data cube is

Fig. 4 Spacecraft configuration (front view)

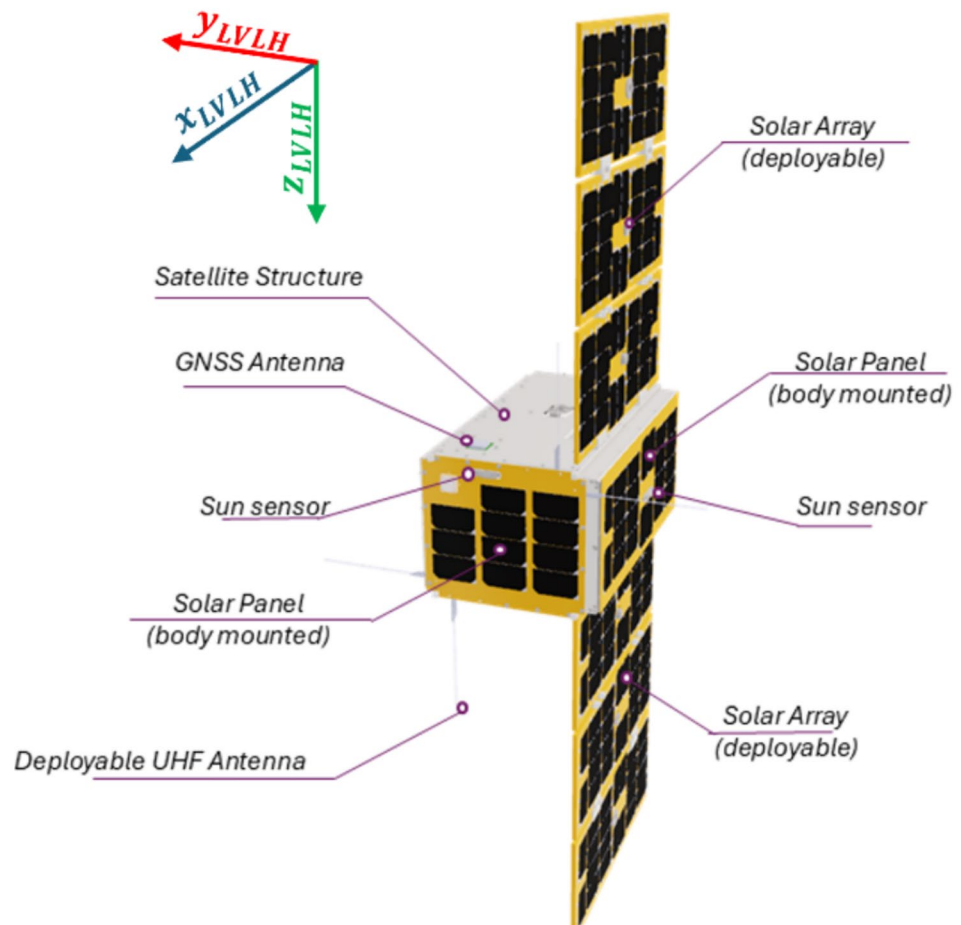


Fig. 5 TMA telescope architecture for EarthNext

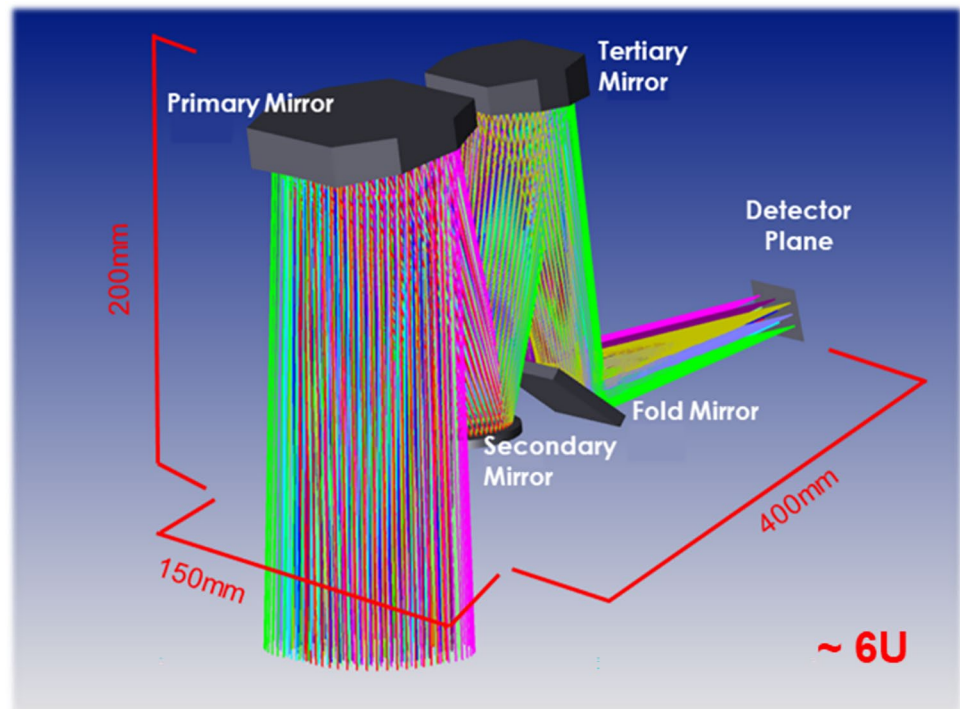


Table 2 Design Parameters of the TMA Telescope

Parameter	Value
Focal Length (mm)	500
Rel. Aperture	F/6
FOV, ACT $\times$ ALT ($^{\circ}$)	3×1.5
Distortion	50%
Pixel size (μm)	6.4
GSD at nadir (m)	3.9
ACT swath width (km)	17

built thanks to the adoption of a multispectral filter array combined with the platform motion, as depicted in Fig. 6.

The acquired payload data result in 2D images at full spatial resolution, comprising a number of spectral stripes for data categorization. To enhance the Signal-To-Noise ratio (SNR), the consortium's analysis recommended adopting both a binning technique and an FMC maneuver, which, in turn, also increases the exposition time over a the AOI.

3.2 Propulsion subsystem

The PS is the REGULUS-50-I2 Large, designed and developed by T4I. It is based on a Microwave Electrothermal Plasma Thruster (MEPT) technology, with a design properly modified to suit CubeSat platforms. The thruster provides a specific impulse up to 550 s and a thrust of 0.5 mN along the thrust direction, used for the drag-compensation maneuver. The subsystem follows a “plug-and-play”

philosophy and is fed with iodine propellant. Its dimensions and weight have a limited impact on the total spacecraft volume and mass. For the sake of completeness, the main characteristics of the thruster are reported in Table 3, while its representation is reported in Fig. 7.

The PS's operational cycle consists of a Propulsion Subsystem Heating (PSH) phase to reach the optimal thruster temperature, a PSA phase and a Propulsion Subsystem Cooling (PSC) phase to bring the system back to safe thermal conditions. We refer to Sect. 4.1 for more details. The PS has been sized to compensate for the drag effects in a timespan between 3 and 5 years. The analyses conducted by the consortium confirmed that the most effective strategy for sizing the spacecraft is to apply a continuous thrust for approximately 2.5 h whenever an average 300 m reduction in semi-major axis reduction is detected. This means that the thruster activation is continuous, but the drag-compensation strategy is quasi-periodic. To make the PS sizing procedure effective, a consistent and iterative study on the expected mean aerodynamic drag value has been carried out, hand in hand with the analysis of the Fire-Off (FO) time, i.e., the time between the end of a PSA and the beginning of the subsequent one. In particular, the objective was to analyze whether the FO time was always sufficient to ensure the correct execution of the PSH and PSC phases, as well as the full energy recovery of the batteries. The core aspect for both the sizing procedure and the FO analysis was the accurate modeling of the aerodynamic deceleration value, typically represented as in (2):

Fig. 6 Spatio-spectral scanning acquisition technique

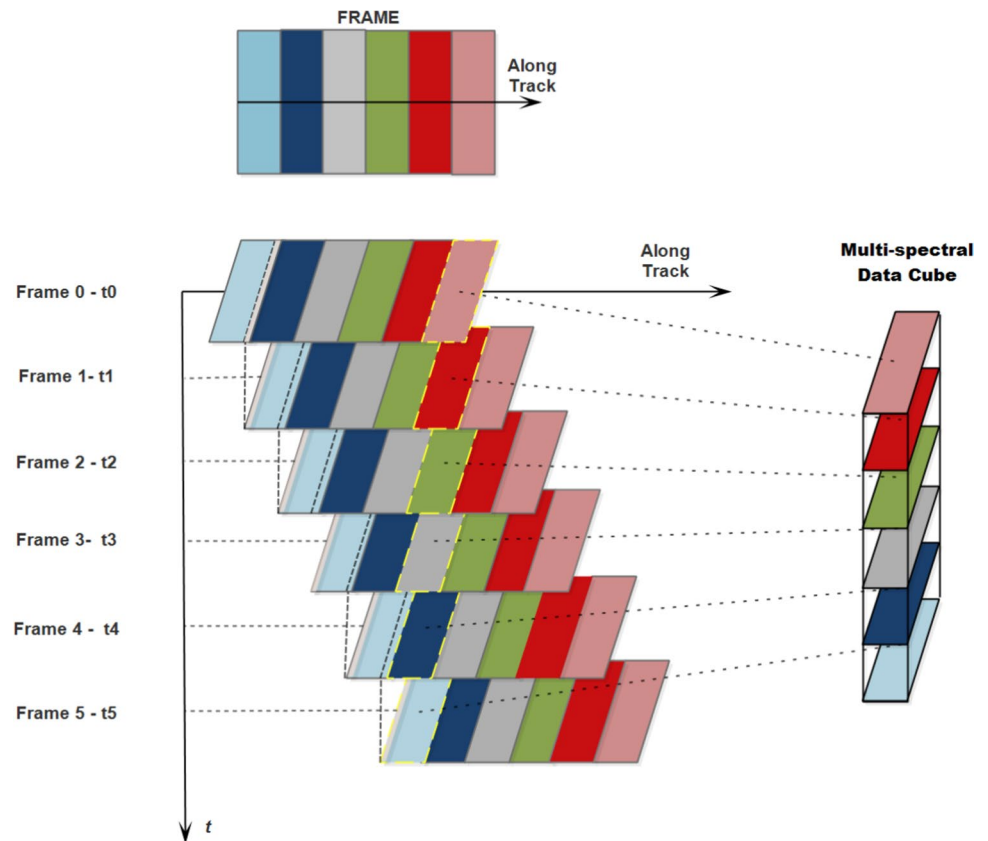


Table 3 PS main characteristics

Parameter	Value
Thrust (mN)	0.1–0.5
Specific impulse (s)	550
Required power (W)	30–50
Mass flow rate (mg/s)	0.1
Propellant	Iodine
Volume (U)	2.2
Wet mass (kg)	~6

$$a_d = -\frac{1}{2}\rho_{\text{atm}}\beta V^2 \quad (2)$$

where a_d is the aerodynamic drag, ρ_{atm} is the atmospheric density, V is the magnitude of the spacecraft velocity with respect to the atmospheric wind, and β is the ballistic coefficient define as in (3):

$$\beta = \frac{C_d A_{\text{ref}}}{m_{\text{sc}}} \quad (3)$$

where C_d is the drag coefficient, A_{ref} is the reference area and m_{sc} is the spacecraft mass. The model used to extrapolate the atmospheric density values at several altitudes is the NRLM-SISE-00, as suggested by the ECSS [12], although the literature acknowledges that the model tends to overestimate

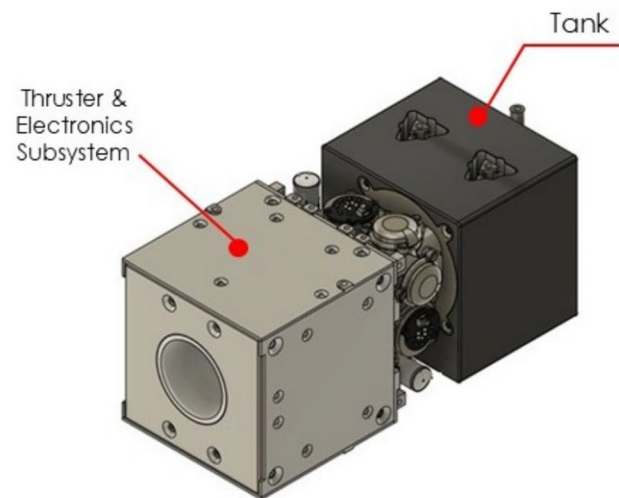


Fig. 7 PS representation

density values compared to in-flight measurements [4, 19]. As is known, the aerodynamic resistance is closely related to the intensification of solar activity, which follows periodic cycles lasting 11 years. As a result, the PS has been sized according to the global WC scenario, identified by considering the mean monthly atmospheric density according to the space weather conditions of the solar cycle 23, during the

2000–2002 triennium, characterized by an average F10.7 value of 181.47 SFU, and an average Ap index of 13.65. Conversely, mission operations, especially in terms of FO times, were subsequently verified through dedicated simulations reproducing the full operational scenario of EarthNext. These simulations employed space weather conditions representative of the statistically WC scenario, as also indicated by the ECSS, and identified during solar cycle 24, over the 2013–2015 triennium. This period was characterized by an average F10.7 index of 128.92 SFU and an average Ap index of 9.20. Once obtained a preliminary PS sizing and verified the possibility to operate with such a PS, the FO time analysis has been improved according to a parametric study based on the range of β values which EarthNext can achieve during the operations and continuously refining the β variation range with the progress of the mission design. Indeed, the first-order analysis was performed using extreme values of the β , derived from minimum/maximum allowable variations in reference surface area and mass (based on deployer constraints), and a WC drag coefficient taken from literature [20, 21]. By contrast, the refined approach utilized the estimation of the normalized drag force factor, defined as in (4):

$$C_{F,A} = C_d A_{\text{ref}} = f\left(s, \alpha, \gamma, \frac{T_w}{T_\infty}\right) \quad (4)$$

which accounts for the azimuth (α) and elevation (γ) configuration of the spacecraft (i.e., the spacecraft attitude with respect the relative speed vector), as well as for the ratio between the wall temperature (T_w) and the free-stream temperature (T_∞) and the speed ratio, in turn defined as in (5):

$$s = \frac{V}{c_\infty} \quad (5)$$

where c_∞ is the most probable molecular speed, in turn influenced by the space weather conditions. In so doing, β can be seen as the ratio of only two quantities, as reported in (6):

$$\beta = \frac{C_{F,A}}{m_{sc}} \quad (6)$$

An analysis carried out by an external consultant with expertise in fluid dynamics characterization highlighted that, within the initially estimated operational triennium (2026–2029) at EarthNext's operational altitudes, the speed ratio is expected to vary from 7.2 to 8.2 with a 90% confidence interval. The expected solar indices for the same period, obtained from the ESA COMSOL database, indicate a mean F10.7 of 106.35 SFU and a mean Ap of 13.16. As an additional level of detail, the drag factor normalized coefficient $C_{F,A}$ was obtained through a panel-method exploiting a gas-surface interaction (GSI) model based on the quasi-specular behavior assumption.

In particular, a Schaaf–Chambre model [22] was used, with both the normal and tangential accommodation coefficient values obtained from a sensitivity analysis on the GSI parameters performed by the external consultant. Based on the maximum and minimum mass configuration of the spacecraft (with the maximum mass of $m_{\text{max}} = 35.6\text{kg}$ in case of full propellant mass plus margin), an aerodynamic database has been built for the expected operational triennium of EarthNext and for the expected range of azimuth and elevation assumed for the spacecraft, identifying the range of atmospheric deceleration force oscillating between an optimistic Best-Case (BC) of 10^{-5}N and pessimistic WC 10^{-4}N . The analysis over the EarthNext operational period confirmed that the PS can sustain the required drag-compensation maneuvers, while consistently preserving sufficient time to perform the PSH and PSC phases at each PSA, even in the conservative case of 90% of the maximum Isp and 90% of the maximum thrust. Additionally, the consortium defined a Propulsion System Temperature Maintenance Mode (PSTM), detailed in Sect. 4.1. This mode is activated when successive PSA phases are required in close succession, such as during periods of intense solar activity. In such cases, the PSH of the second cycle and PSC phase of the first cycle are bypassed; instead, a continuous power input is used to maintain the thruster temperature at a ready-to-fire level. This approach has also been verified from an electric power-budget perspective and deemed feasible. Finally, the PS was sized to provide an overall $\Delta V > 275\text{ m/s}$, ensuring at least 3 years of continuous operations. Due to volume constraints, the thrust direction is, however, misaligned with respect to the center of mass, producing a disturbance torque that has become a driving requirement for the sizing of attitude-control actuators. To clarify this aspect and complement Fig. 4, the back view of the spacecraft is shown in Fig. 8.

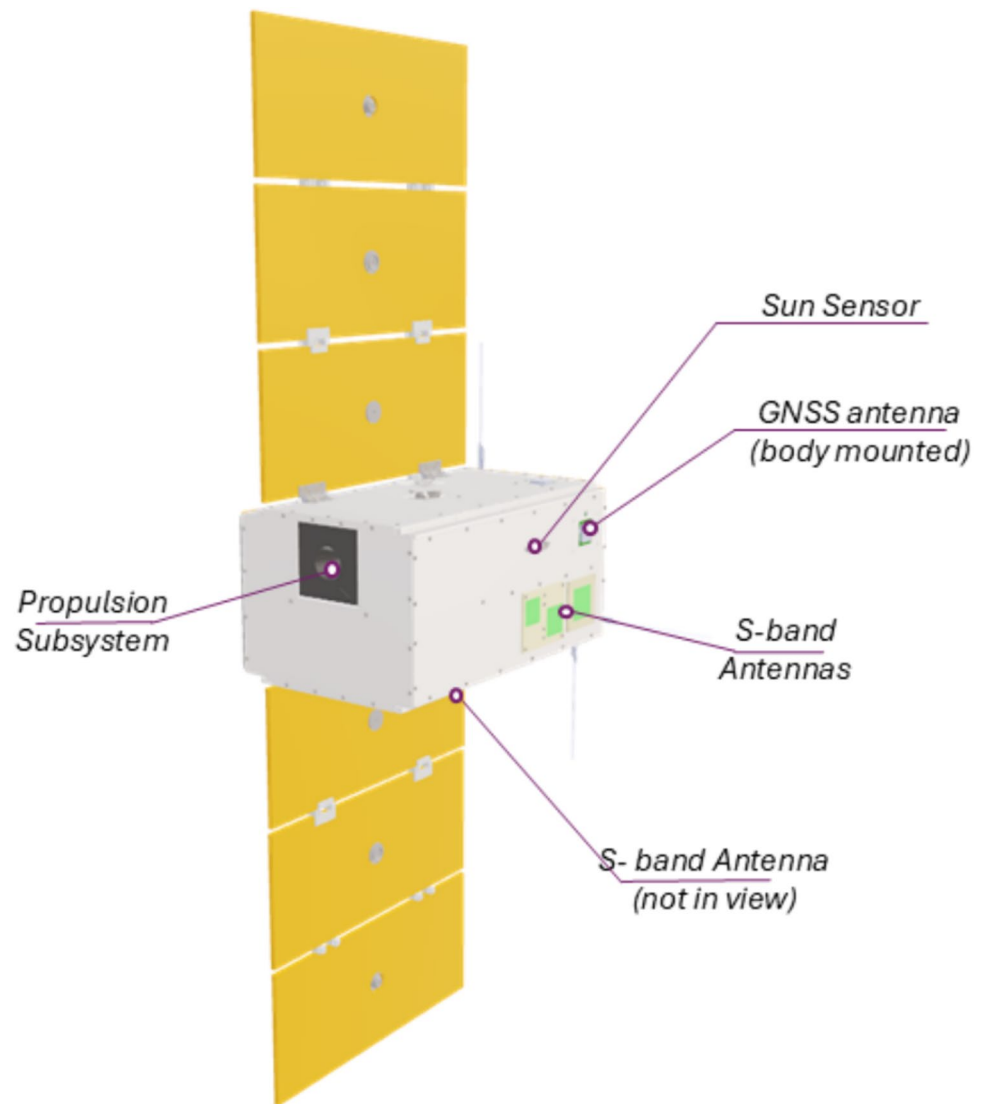
The thruster configuration also prevents the plume impingement over external parts of the spacecraft, as further confirmed by the team's analysis about the plume assessment. Figure 9 depicts the plume cone expected from the thruster.

3.3 GNC subsystem

The sensing part of the GNC subsystem includes:

- #1 star tracker;
- #3 sun sensors;
- #3 MEMS magnetometers;
- #1 MEMS Inertial Measurement Unit (IMU);
- #2 GNSS Antennas.

Fig. 8 Spacecraft configuration (back view)



The single star tracker has been placed on the top part of the spacecraft to ensure good and continuous visibility of the sky. Its tilted configuration has been selected based on its FOV, the presence of the solar-panel occultation, and on the Earth/Glint keep-out zone information provided by the constructor. Two sun sensors are provided on the lateral sides of the spacecraft main body, while a third sun sensor has been placed on the front of the spacecraft, to optimize the sun-detection capabilities. A schematic of the configuration of the star tracker, of the sun sensors and of the secondary camera for the cloud detection has been reported for the sake of clarity in Fig. 10.

The provided sensor configuration ensures an Absolute Knowledge Error (AKE) smaller than 0.01° during the APs, even considering the expected WC FMC of $1^\circ/\text{s}$ around the pitch axis and the expected WC GNSS position estimation. During the other mission phases, the AKE requirement is less than 1° for the detumbling phase and

less than 0.1° for all other phases, with which the platform remains compliant. Future development activities will include an AKE analysis considering transitions between sunlit and eclipse conditions.

As regards the control hardware, the spacecraft is equipped with:

- #4 Reaction Wheels (RWs) in a skewed custom configuration. Each RW provides a maximum torque of 0.007 Nm and a maximum angular momentum of 0.1 Nms;
- #3 torque rods, each one aligned with one of the axes of the BRF, providing a magnetic dipole of at least 2.6 Am^2 , sized so to perform initial detumbling and RW desaturation.

The configuration of the control-hardware, able to provide an Absolute Performance Error lower than 0.1° during

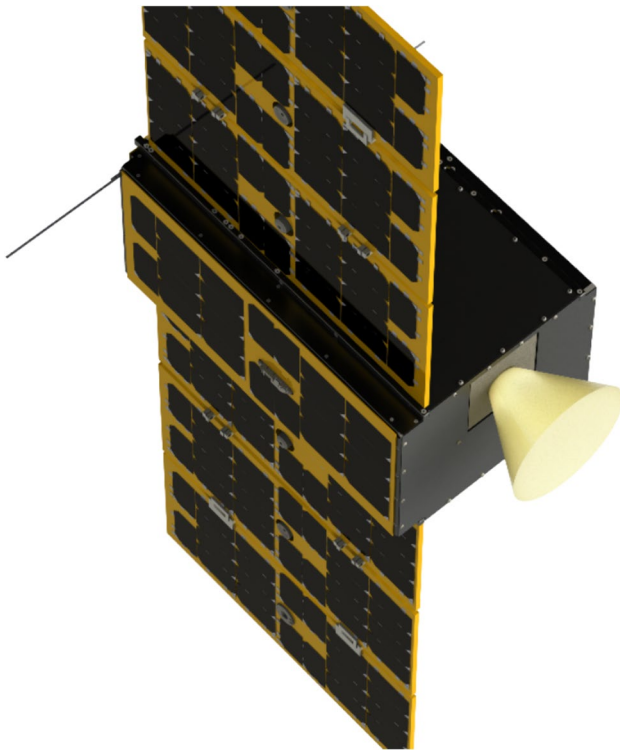


Fig. 9 Plume assessment representation

APs, is provided in Fig. 11, separating the RWs configuration from the torque rods one.

Note that in Fig. 11, the term BRF denotes the Body Reference Frame, whose axes are aligned according to the spacecraft principal axes of inertia and centered in the spacecraft Center of Gravity (CoG). Conversely, $\vec{t}_{RW_i}$ is the

i -th wheel torque, while $\vec{m}_i$ is the i -th torque rod magnetic dipole. The RWs sizing procedure was mainly driven by the disturbance caused by the PS thrust misalignment, as it acts for 2.5 h and being the largest in magnitude. For the preliminary selection of the hardware at the beginning of the Phase B, it was assumed a max thrust disturbance torque in the order of 10^{-5} Nm on two of the three body axis (the x_{BRF} was excluded), along with an in-plane drag torque component in the order of 10^{-6} Nm. The analysis highlighted the impossibility of using only 3 RWs due to a negative momentum budget on one of the wheels. For this reason, a four-reaction-wheels configuration with a customized skewed layout was selected. In Table 4, according to the last mission-design updates, the expected maximum order of magnitude of the main environmental disturbance torques as well as the PS disturbance torque are reported accounting for all the attitude maneuvers planned for EarthNext. The mean value is computed on the timeframe of PSA, since considered as the hardware sizing phase and because off the RWs desaturation phases which occurs before and after each PSA. The aerodynamic torque has been evaluated following the same methods and hypotheses explained in Sect. 3.2 for the aerodynamic deceleration characterization. For such analysis, the aerodynamic torque is evaluated, per each axis, as reported in (7)

$$M_i = \frac{1}{2} \rho_{\text{atm}} V^2 C_{M_i} A_{\text{ref}} L_{\text{ref}} = \frac{1}{2} \rho_{\text{atm}} V^2 C_{M_i, A, L} \quad (7)$$

where $i = x_{BRF}, y_{BRF}, z_{BRF}$, C_{M_i} is the momentum coefficient related to the i -th axis, L_{ref} is the reference length, and $C_{M_i, A, L}$ is the normalized aerodynamic momentum coefficient,

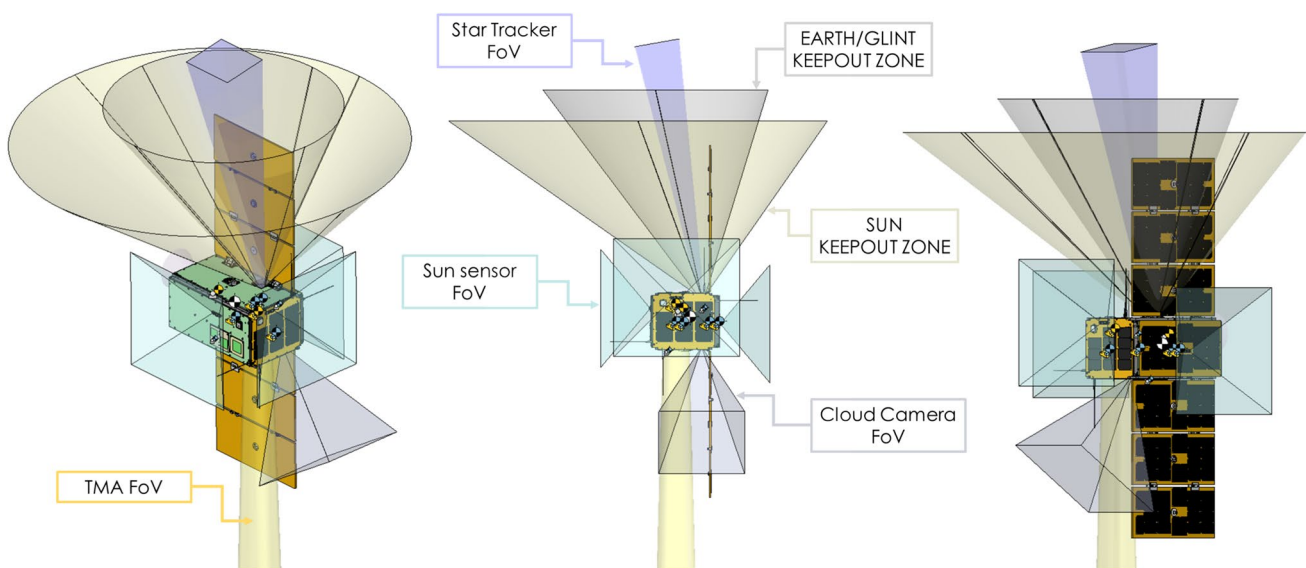
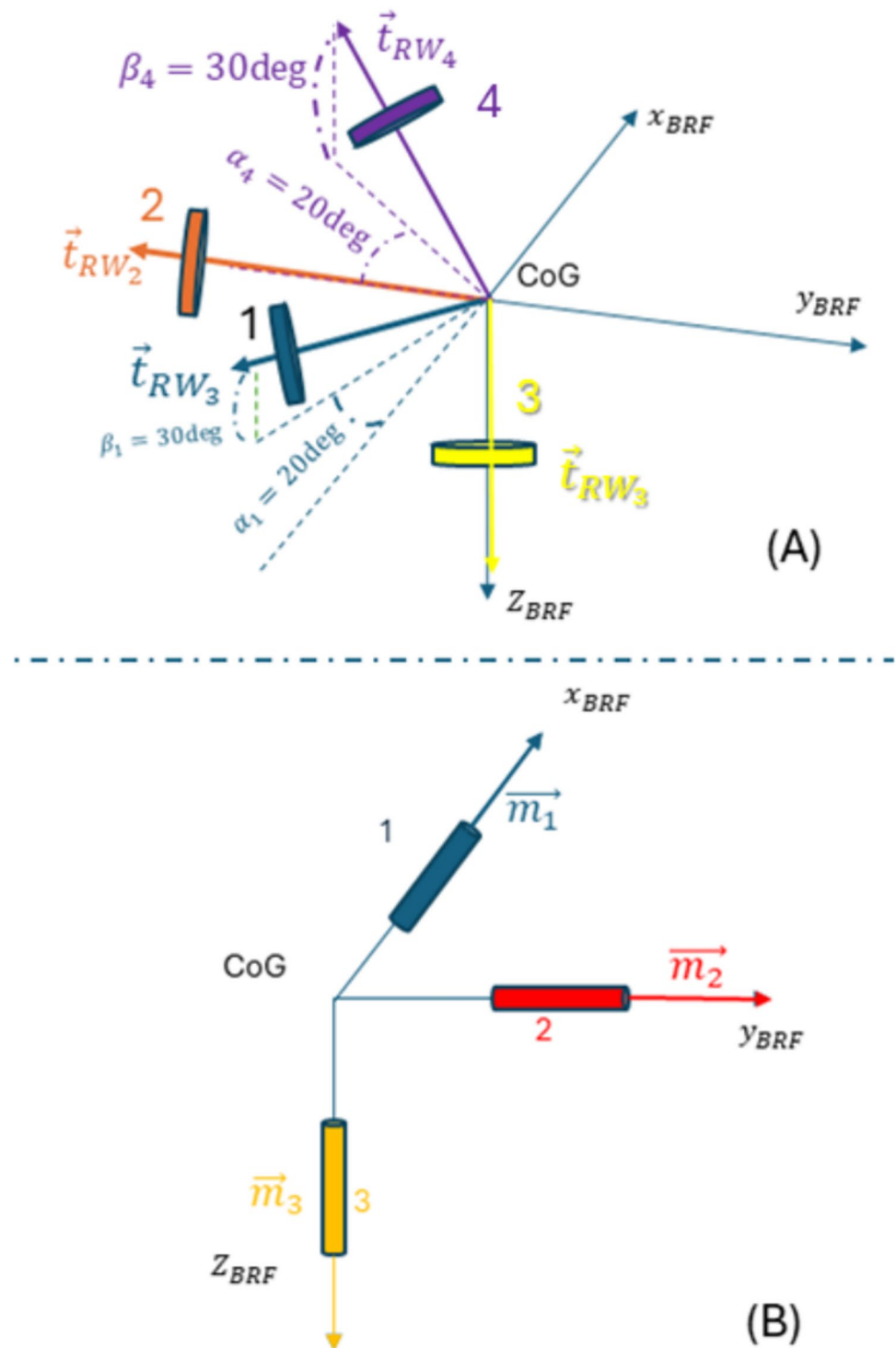


Fig. 10 Attitude-determination sensors and secondary-camera configuration

Fig. 11 Control hardware configuration. (A) The RWs assembly is depicted; (B) the torque rods configuration has been reported



whose database has been obtained as function of azimuth, elevation and speed ratio of the spacecraft. The gravity-gradient has been modeled according to the well-known relation reported in [23]. The maximum recorded value is reasonably obtained when the spacecraft is in roll-bias mode. Due to the persistent roll bias throughout the entire PSA phase, the gravity gradient torque will remain continuously active

around the computed value. The magnetic residual-dipole torque, modeled as in [23], has been evaluated according to two different residual magnetic dipole per axis scenarios, respectively equal to 0.1 Am^2 and 0.01 Am^2 , using the IGRF-13 as the magnetic field model, the last updated version available at the time of simulations. The results suggest the need to test the electromagnetic compatibility of

Table 4 Estimated disturbance torques order of magnitude

Disturbance torque	Max. order of magnitude Nm	Mean order of magnitude during PSA Nm
Thrust misalignment	10^{-5}	10^{-5}
Aerodynamic torque WC	10^{-5} (during AP)	10^{-6}
Aerodynamic torque BC	10^{-6} (during AP)	10^{-6}
Gravity gradient	10^{-6} (during PSA)	10^{-6}
Magnetic residual dipole ($m_i = 0.1 \text{ Am}^2$)	10^{-5}	10^{-6}
Magnetic residual dipole ($m_i = 0.01 \text{ Am}^2$)	10^{-6}	10^{-7}
Solar radiation pressure	10^{-7}	10^{-8}

the spacecraft, since the resulting disturbance may be on the same order of magnitude as the aerodynamic one. It is here highlighted that the electromagnetic-compatibility tests for the spacecraft are required also because of GNC hardware constraints. Finally, a panel-method has been employed for the evaluation of the solar radiation pressure torque, following the procedure reported in [24] in the WC assumption that all the surfaces have the maximum reflective coefficient. The RWs skewed configuration has been instead optimized to counteract the sum of all the disturbance torques acting on each axis of the spacecraft BRF, guaranteeing compliance with the mission attitude-pointing stability requirements during all mission phases.

3.4 AI and automation

The EarthNext project envisages implementing AI-based logic to:

- perform precision agriculture or to perform change detection over the AOI via segmentation process, identifying changes in the map;
- process acquired images and extract only customer-selected features, using a data-prioritization logic without the need of ground commanding;
- perform a real-time telemetry analysis for prognostics and diagnostics of the spacecraft's status;
- perform the online replanning of the schedule of operations, based on the prediction about the space weather conditions.

Additionally, the AI-based logics allows to optimize the data handling by merging the information on overflight

opportunities with the information coming from the cloud camera, providing:

- an optimization of the observation events taking cloud coverage into account to prioritize clear-sky acquisitions;
- an optimization of the memory management, allowing to stow on-board all the useful information and send them to ground when efficiently.

The AI-based online replanning capability relies on a drag estimation pipeline based on the prediction of space weather indices (F10.7 and Ap), an operations ICD incorporating time constraints and task priorities, and the integration of an orbital propagator with a cloud detection module validated on the designated on-board platform.

3.5 Payload data transmission and TT&C subsystem

The payload-data transmission and TT&C subsystems of EarthNext rely on:

- #1 X-band antenna for the download of acquired EO related data (Tx Data rate = 150 Mbs);
- #2 S-band antennas for TT (Tx Data rate = 2 Mbps, Rx Data Rate = 112 kbps);
- #2 four-in-one deployable UHF antenna for TT (Tx/Rx Data Rate > 9 kbps).

Such a configuration provides EarthNext with a reliable communication subsystem to interface with ground facilities, supporting both high-volume payload-data downlink and relatively large T&T data volumes. The baseline data volume assumed for the analysis corresponds to an uninterrupted 8-min overflight, totaling 51.8 Gbit of acquired data, representing a non-realistic WC scenario based on the maximum distance the spacecraft would fly over the AOI. Analyses conducted under the same operational constraints described in Sect. 2 indicate that at least two ground station GSs contact windows are required to complete the data downlink. A maximum of three contacts may be needed only in the unlikely scenario where all three available GSs contacts between two consecutive overflights result in minimum-duration visibility windows. At this stage of the mission design, the telemetry data volume has not yet been formally defined. However, a preliminary estimate of the potential Tx/Rx capacity via S-band communication for the selected hardware has been made, which, according to the consortium's experience, is well in excess of the expected requirements.

3.6 Electrical power subsystem

As shown in Fig. 4, EarthNext is equipped with:

- #6 deployable-panel, symmetrically distributed about the spacecraft main body, which forms the main solar array. Its configuration is optimized for the SSO configuration to ensure always good illumination conditions when possible;
- #1 side-panel body mounted;
- #1 frontal panel body-mounted, of reduced dimension, to accommodate its shape on the frontal area of EarthNext;
- a battery pack in a 4S6P configuration.

The number of solar panels has been selected to ensure full battery recharge and proper platform operation, even in the event of a single-string fault under EOL conditions for both the solar panels and batteries. Considering the orbital configuration, a roll bias maneuver during the PSA phases maximizes the electrical power harvested by the solar array. Finally, it should also be noted that, due to the quasi-SSO characteristics, the optimal roll bias to maximize the generated electrical power may vary over the mission lifetime, according to the LTDN variation range (but proved to be still affordable by the GNC subsystem).

4 Concept of operations and GNC modes

The EarthNext baseline mission profile comprises several phases and GNC operation modes. The EarthNext mission phases represent the chronological and logical sequence of mission operations. In the following, the main mission phases have been briefly described:

- *Launch and Early Orbit Phase (LEOP)*: this phase starts as soon as the satellite is released from the deployer. The spacecraft will go first into a low-functioning state (activity of onboard equipment will be minimal and limited to the necessary functions) in which the EPS, Payload and Platform Data Subsystem (PPDS) and GNC subsystem are powered on and checked out to verify basic functions and health status. A detumbling phase is then executed thanks to the torque rods. Once the desired angular velocity is reached, the UHF antennas are subsequently deployed;
- *Subsystems Commissioning Phase @360 km (S/S CP#1)*: after reducing the satellite's angular rate and achieving the desired attitude, the solar panels are deployed. The spacecraft then begins the commissioning phase, during which all major subsystems (excluding the payload) are powered on and tested individually. While some checks are automated, the process is largely monitored and controlled by the ground segment, with subsystem leads providing immediate feedback. This phase follows a detailed pre-established commissioning plan;
- *Payload Commissioning Phase @360 km (P/L CP#1)*: once achieved the nominal attitude, the commissioning activities for the EarthNext EO payload will be carried out. In particular, the checking of the full operability of the payload is performed. A detailed characterization and calibration/validation procedures will be carried out to fine-tune and validate the payload's performance at this altitude;
- *Nominal Operation Phase #1 (NOP #1)*: this phase corresponds to a nominal operative phase of the mission, further described in Sect. 4.1 with all its sub-modes. During this phase the electro-optical payload performs its measurements and science data are downlinked to GSs. The activity of the spacecraft, its attitude and position and its status are monitored by telemetry data that are regularly transmitted to ground. The typical onboard operating modes of EarthNext are envisaged as fully automated, with a set schedule defined in the software for the duration of each mode and the timing/conditions for transfer between them. Nevertheless, ground control will have the ability to switch the satellite between modes and override the planned schedule as desired/required for proper satellite function. During NOP#1 however, only one complete operations cycle will be performed, to test the spacecraft ability to complete the mission objectives at that altitude;
- *Drag-aided transfer to target orbit phase (DA-TTOP)*: if in the NOP #1 the orbit maintenance is guaranteed as expected, the DA-TTOP phase starts. It implies a natural decay of the orbit until the target orbit at 302 km. For the sake of safety, during the natural decay, checkpoints are planned to test the PS capability to perform drag compensation. Up to 320 km of altitude, four check-points at a distance of 10 km in altitude will be planned. Below 320 km of altitude, at least three checkpoints will be planned, with a max altitude distance of about 5 km each-one. The maximum threshold in altitude spacing has been preliminary identified according to a team analysis about the expected WC environmental scenario and the capability of the thruster to compensate for the 300 m altitude decay. However, the checkpoints are not fixed and can be replanned according to the prediction on the weather conditions and platform status. A batch of drag-aided transfer scenarios was analyzed by the consortium, using Space Weather conditions from the 2013–2015 triennium, seven NOP checkpoints, and assuming no attitude variation to increase the wet surface, leading to an estimated maximum transfer time of six months.

Once reached the nominal operation orbit, the following mission phases are envisaged:

- *Payload Commissioning Phase @302 km (P/L CP #2)*: which is characterized by the same operations as the P/L CP @360 km but performed at different altitude;
- *Nominal Operation Phase #2 (NOP #2)*: Equivalent to NOP#1, but at 302 km, i.e., the nominal operative scenario continuously repeated until the end of the mission;
- *Decommissioning Phase (DP)*: No specific requirement is still issued for the disposal phase. At EOL, due to drag effects, the satellite will naturally re-enter the atmosphere in compliance with the current regulation for debris mitigation and burn up completely.

A pictorial view of the Concept of Operations for EarthNext is reproposed in Fig. 12.

Fig. 12 Pictorial view of the concept of operations for EarthNext

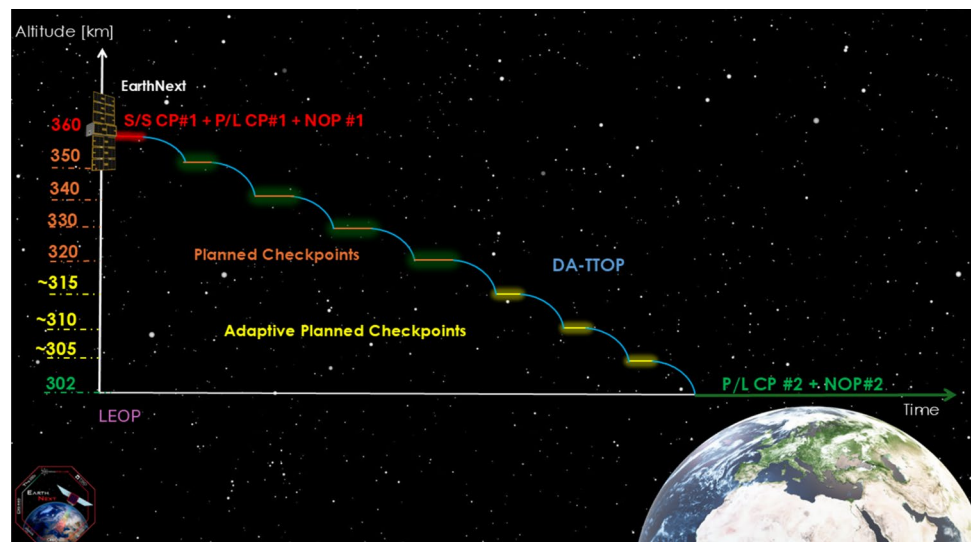
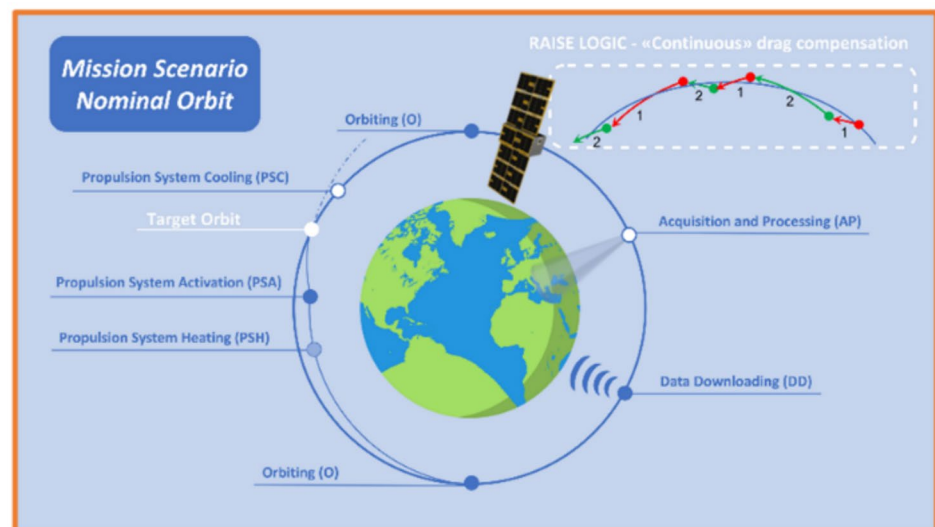


Fig. 13 NOP pictorial representation



4.1 The nominal operation phase

The NOP depicted in Fig. 13, includes the following operational sub-modes:

- *Acquisition and processing (AP)*: the payload collects high-resolution images and the platform performs the attitude maneuvers necessary for achieving the desired product level;
- *Propulsion subsystem heating (PSH)*: a preheating action for the proper working of the PS is performed, the goal is to reach a specific PS temperature to optimize the thrust performance;
- *Propulsion subsystem activation (PSA)*: the PS is activated for the duration required to restore the target orbit altitude. The system is sized to compensate for every

300 m of reduction in mean SMA, resulting from an optimized maneuvering strategy for the drag compensation;

- *Propulsion system temperature maintenance mode* (PSTM): this mode is activated when operational constraints call for closely spaced subsequent PSA phases to maintain the desired altitude, in case of intense solar activity. This means that the PSH preceding the second PSA and the PSC following the first PSA are no longer required for the subsequent firings; instead, a continuous power input is used to maintain the thruster temperature at a fixed level, ready-to-fire;
- *Propulsion subsystem cooling* (PSC): this mode is a controlled cooldown mode, before turning off the PS. It is not required when the PS must be reactivated within a short time (PSTM);
- *Data downloading* (DD): when a communication window is available and does not fall in a PSA or PSH, the data acquired during AP are transferred to the network of ground-stations, without performing any attitude maneuver;
- *Orbiting* (O): this mode can be ascribed as an idle state of the CubeSat. The primary objective of the spacecraft is to ensure a positive power budget to maintain the state of charge of its batteries, recharging as needed.

4.2 The GNC mode during NOP

To conclude, a brief description of the NOP-related GNC modes is provided to better frame the attitude maneuvers the spacecraft will perform. In particular, NOP encompasses the following GNC modes:

- *Payload acquisition mode* (PAM): during the APs with the payload, the spacecraft performs an FMC maneuver to compensate for orbital motion. The sensor suite is expected to be fully operative, while the RWs shall ensure the prescribed pointing accuracy;
- *Roll-bias mode* (RBM): before (PSH), during the PSAs, and after (PSC), a roll-bias attitude configuration shall be achieved to maximize the available electric power and reduce the depth of discharge of the batteries. The roll-bias is also kept during the whole PSTM. The attitude and orbit determination sensors suite is expected to be fully operative, while the RWs shall guarantee the prescribed accuracy;
- *Velocity pointing mode* (VPM): during PSA, the spacecraft shall guarantee both a roll-bias configuration for the power management, as well as thrust direction alignment parallel to the direction of the velocity vector within pre-described attitude requirements;
- *Desaturation mode* (DM): before and after the PSAs (or in case of prolonged O), desaturation events are

required to reduce the angular momentum of the RWs. The desaturation ends when a safe and minimum angular momentum level is reached for each RW.

5 Conclusions

The following contribution presents the recent progress in the EarthNext mission and platform design. The spacecraft is intended as an in-orbit demonstrator to exploit CubeSat in VLEO, at an equatorial mean altitude of 302 km. The nominal mission orbit has been defined according to the mission scientific objectives, i.e., performing customer-based multispectral acquisitions over Italy. The spacecraft design foresees the use of two enabling technologies, which are an innovative, CubeSat-sized, electric propulsion subsystem and an imaging system based on a three-mirror anastigmatic telescope. Furthermore, the design also envisages the onboard autonomous decision-making and operations-planning capabilities, thanks to AI-based algorithms. The platform, characterized by a robust and agile GNC subsystem, can guarantee the required pointing stability during the data acquisition while compensating for strong environmental and platform-related perturbations. The upcoming mission phases will focus on consolidating the design choices, based on detailed analyses of the environmental characterization and the integration of the various satellite subsystems. Key challenges will include achieving the highest possible accuracy in the modeling of aerodynamic disturbances, conducting electromagnetic compatibility testing of the spacecraft, performing structural and integration tests, and implementing and validating AI-based optimization logic for mission operations. Specifically, a new methodology will be developed for estimating and comparing the expected aerodynamic disturbances with real-time measurements, while a novel procedure will be introduced for AI-based optimization, marking the first implementation of AI-driven logic on a VLEO satellite.

Author contributions Tobia Armando La Marca wrote the full paper.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare that there are no competing interests.

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